

Epidemic Models in Complex Networks

The study of epidemic spreading in complex networks is crucial for understanding the mechanisms governing the persistence or extinction of infectious diseases in populations. Beyond epidemiology, these models also serve as paradigmatic examples of non-equilibrium phase transitions and intrinsic universal criticality in statistical physics. Among the various models used to describe epidemic spreading, the *Susceptible-Infected-Susceptible* (SIS) model stands out as fundamental framework. The SIS model (which is directly related with the Contact Process model) describes infections where individuals recover but do not acquire immunity, allowing for reinfection. This model is particularly relevant for diseases such as the common cold or hospital-acquired infections, where immunity is short-lived or nonexistent.

In this project, we will explore how the SIS model critical behaviour is affected by either heterogeneity effects or the addition of new classes of individuals. Consequently, the study is split into two complementary works.

Topology effects on the SIS model

A defining feature of the SIS model is the presence of a critical transition. When the infection rate is below a certain threshold, the disease eventually dies out; above this threshold, it persists indefinitely in the population. This transition belongs to the directed percolation universality class, a fundamental concept in statistical mechanics. The critical point is not only an important epidemiological feature but also a key signature of the intrinsic universal criticality of this model. In heterogeneous networks, such as those found in real-world social interactions, the epidemic threshold exhibits complex behavior that deviates significantly from classical mean-field predictions.

This work aims to explore the critical transition in the SIS model, analytically and computationally, while also considering how network topology influences the epidemic threshold.

- **First task:** (*Easy*) The SIS model can be described using a mean-field approximation, which assumes that individuals interact in a well-mixed environment. Under this assumption, the fraction of infected individuals $\rho(t)$ evolves according to the equation:

$$\frac{d\rho}{dt} = \beta\rho(1 - \rho) - \mu\rho, \quad (1)$$

where β represents the infection rate and μ is the recovery rate. The steady-state solution of this equation reveals the presence of a critical infection rate β_c , below which the disease dies out. Derive that mean-field equation from the Master Equation analysis.

- **Second task:** (*Intermediate*): A more realistic approach involves considering the *degree-based mean-field approximation*, which accounts for network heterogeneity. Derive the mean-field equation considering this new approach. By solving for the steady-state condition, show that the epidemic threshold in heterogeneous networks follows:

$$\lambda_c \sim \frac{\langle k \rangle}{\langle k^2 \rangle}. \quad (2)$$

- **Third task:** (*Intermediate*): To numerically investigate the epidemic threshold, one can implement a simple discrete-time algorithm to simulate the SIS model on different network topologies, such as Erdős-Rényi (random graphs), Watts-Strogatz (small-world networks), and Barabási-Albert (scale-free networks). By measuring the fraction of infected individuals ρ over time and identifying the infection rate β at which the disease transitions from extinction to persistence, the epidemic threshold can be determined.

To do that, use the *susceptibility* of the system, given by

$$\chi = N (\langle \rho^2 \rangle - \langle \rho \rangle^2), \quad (3)$$

as an order parameter. It exhibits a peak at the transition point. Near the critical threshold, the prevalence follows a power-law scaling,

$$\rho \sim (\beta - \beta_c)^\beta, \quad (4)$$

where the exponent β is characteristic of the directed percolation universality class. By varying the network size, it is possible to determine the finite-size scaling exponents and confirm the universal behavior of the transition.

- **Fourth task:** (*Hard*): An additional and/or alternative extension to the model consists on a heterogeneous infection rate, which may differ for each node. This modification can lead to the emergence of a huge region of critical dynamics, known as *Griffiths phase*. Study the emergence of this region in the SIS model.